WORKSHEET

Chapter 8: Related rates

1 Find an expression for $\frac{dy}{dt}$ given:

a $y = 3x^2 - 4x$ and $\frac{dx}{dt} = 2$

b $y = 4e^{3x} + 7$ and $\frac{dx}{dt} = 5$

c $y = 4\cos(x) + 2x^2$ and $\frac{dx}{dt} = \frac{1}{2}$

d $y = 3x^3 + 2x - 7$ and $\frac{dx}{dt} = -2$

2 Evaluate $\frac{dy}{dt}$ at $x = 3$, given:

a $y = 3x^3 + 2x - 7$ and $\frac{dx}{dt} = 4$

b $y = (2x - 1)^3$ and $\frac{dx}{dt} = -2$

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c $y = x\sin(x)$ and $\frac{dx}{dt} = 2$

d $y = 3xe^{x^2}$ and $\frac{dx}{dt} = -\frac{1}{2}$

3 If $y = \cos(3x)$ and $\frac{dx}{dt} = 4$, evaluate $\frac{dy}{dt}$ when $x = \frac{\pi}{12}$.

4 A rectangular prism has a base measuring 2 m by 3 m. The prism is being filled with liquid at the rate of $0.0054\ \text{m}^3\ \text{s}^{-1}$. At what rate is the height of the liquid in the prism increasing?

5 A spherical balloon is being filled at a rate of $20\ \text{cm}^3\ \text{min}^{-1}$. How fast is the radius of the balloon increasing when the radius is 6 cm?

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6 A 25 m long steel tube leans against a building. If the bottom of the tube gradually slides away from the building horizontally at a rate of 3 m min^{-1}, how fast is the other end of the tube sliding down the building when the top of the tube is 7 m from the base of the building?

7 A cyclist is travelling on a straight road running due south at 16 km h^{-1}. At the same time, another cyclist leaves an intersection with this road and travels due east at 12 km h^{-1}. At what rate are the cyclists approaching or separating after 1 h?

8 Air is escaping from a spherical balloon at the rate of 2 cm^3 s^{-1}. How fast is the surface area of the balloon shrinking when the radius is 12 cm?

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9 Two sides of a triangle have fixed lengths measuring 12 cm and 15 cm. The angle between these two sides is increasing at the rate of 2° per minute. How fast is the area of the triangle increasing when the angle between the sides is 60°?

10 Water is leaking out of a conical funnel at the rate of 1 $cm^3\ s^{-1}$. If the radius of the base of the funnel is 4 cm and the height is 8 cm, find the rate at which the water is dropping when it is 2 cm from the top of the funnel.

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Answers

1 **a** $12x - 8$

b $60e^{3x}$

c $2x - 2\sin(x)$

d $-18x^2 - 4$

2 **a** 584

b -588

c -5.658

d $\dfrac{-57e^9}{2}$

3 $-6\sqrt{2}$

4 0.9 cm s^{-1}

5 About 0.0442 cm min^{-1}.

6 $\dfrac{7}{8}$ m min^{-1}

7 They are approaching at 5.6 km h^{-1}.

8 $\dfrac{1}{3}$ cm s^{-1}

9 About 1.575 cm^2 min^{-1}.

10 $\dfrac{1}{9\pi}$ cm s^{-1}

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